

Requirement 1

To complete the lab you have to do the following:

- In two different terminals, run `roscore` in the first one and `roslaunch turtlesim turtlesim_node` in the second one
- Open another terminal, and use `rostopic list` & `rostopic <topic_name> info` to get the publishers and subscribers of any topic related to the turtle package



`rostopic <topic_name> info` shows the publishers and subscribers of a specific topic, and also shows the message type used in the topic. If the package is related to the turtle package, you will see something like this:

```
Type: turtlesim/Color

Publishers:
* /turtlesim (http://Ahmeds-MacBook-Pro-3.local:49553/)

Subscribers: None
```



In this topic, `/turtlesim` node is the only publisher, no subscribers, and the message type used is `turtlesim/Color`.

- Which topic has `/turtlesim` node as subscriber? and what is the message type of this topic?
- Create a workspace folder, and then create a package named `turtle_req` with the following dependencies:

- 1- rospy
 - 2- std_msgs and/or geometry_msgs depending on your needs (Hint: See the message type in the previous requirement)
- Create a script named `move_turtle.py` that does the following:
 - 1- Initialize the x component of linear velocity with **0.5**
 - 2- Assign the node as publisher in the topic you found `/turtlesim` node as subscriber in it (In the first question written in **bold** above)
 - 3- For every **2.5** seconds:
 - toggle between making the robot moves horizontally (In the x-component) and vertically (In the y-component) with velocity **0.5**
 - Make a `move.launch` file that runs the following:
 - The node `turtlesim_node` from the `turtlesim` package
 - The node `move_turtle` from the `turtle_req` package (The script you implemented earlier)
 - After launching the file using `roslaunch` command, you should see the following after a while:

